Binary cubic descent: explicit quartics and a CM exclusion test

2026-10-25

The Rust census finds **33 smooth plane quartics over F8** with Jacobian Weil polynomial $T^6 - tT^3 + 512$, for $t = \pm 35$. These are two source trace classes absent from the earlier local spectral journal. They are **not established globally novel affected curves**: the quartic family is published, an evaluated source-to-Jacobian map is missing, and publication priority of the derived criterion remains open. Beating rho is not the acceptance gate.

Exact census and controls

The source model is $E_a : y^2 + xy = x^3 + ax^2 + 59$ over $K = \mathbb{F}_2[z]/(z^9 + z + 1)$, modulus 515, with a=0 or 1. The coefficient 59 has full absolute degree nine. The base field is $k = \mathbb{F}_2[u]/(u^3 + u + 1)$, modulus 11. Integers denote power-basis bitmasks; they are not integer field scalars. No subgroup generator has been selected, so no EC1 subgroup identity or verified ISO1 edge is assigned.

We enumerate the published ordinary O3 normal family of Nart–Ritzenthaler, section 1.4. Write $Q = ax^2 + by^2 + cz^2 + dxy + eyz + fzx$. Here s=2, t=6 and the model is

$$Q^2 = (x + y + z)\{2(x^3 + y^3 + z^3) + xyz + 7(xy^2 + x^2z + yz^2) + 5(xz^2 + x^2y + y^2z)\}.$$

The paper's three nonsingularity conditions and rational C3 coefficient action are applied exactly. In particular $Q(1, 1, 1) \neq 1$ is required. The norm expansion is checked on all 512 base-field triples. Smoothness conditions are compared with the seven split Fano points on 64 binary coefficient vectors. Four fixed models check the optimized point counter against a raw polynomial-arithmetic counter with a different projective normalization. All 33 retained targets are independently replayed over F8, F64 and F512; the two displayed examples also have standalone CLI receipts. This is a second arithmetic path within the same Rust implementation, not an external implementation review.

Stage	Exact number
All coefficient vectors	262,144
Smooth vectors	228,481
Rational C3 orbit representatives	76,189
Representatives with N1=9	21,652
Also N2=65	3,799
Trace -35 targets	30
Trace +35 targets	3

For genus three, the first three point counts determine the Weil polynomial by Newton identities. With these targets it equals the restriction polynomial of the displayed source. Tate's theorem therefore gives an isogeny-class relation. It does not supply an efficient homomorphism or a transferred log.

Source	Example Q coefficient vector	N1,N2,N3	Source and Jacobian order
a=0, trace -35	(5,7,7,6,0,0)	9,65,618	548 = 4 times 137
a=1, trace +35	(3,7,7,4,4,0)	9,65,408	478 = 2 times 239

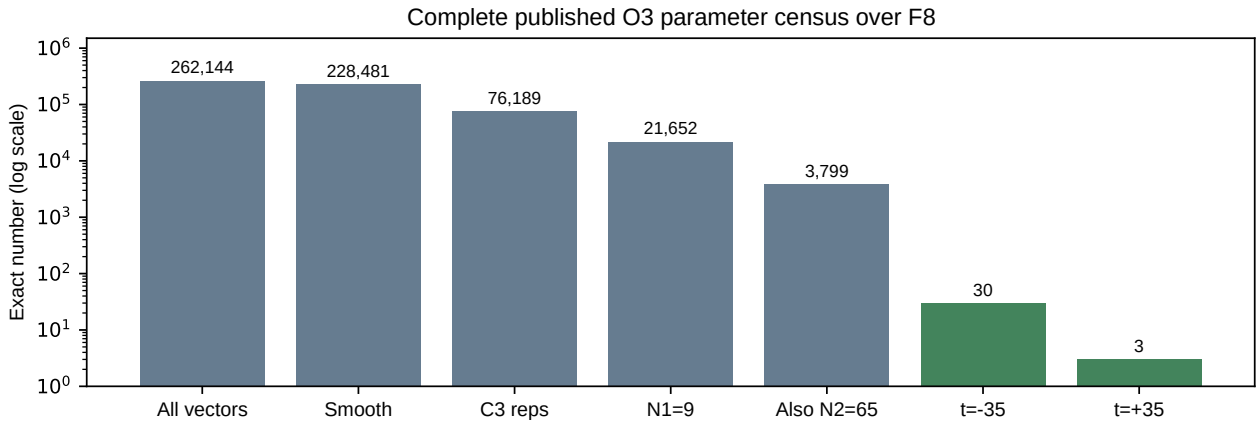


Figure 1: Exact finite census. Bars show counts, with a logarithmic axis; no sampling uncertainty or timing claim.

The JSON standalone replays include the fully expanded quartic polynomials, field embeddings and counts. These coefficients identify equations, not selected prime-subgroup instances.

Geometric automorphisms separate the targets

Conjugating to the split Fano-bitangent model reduces the full geometric automorphism search to all 168 elements of GL3(F2). Every retained matrix is replayed in the original coordinates by polynomial substitution up to scalar; group closure and Frobenius closure are checked separately.

Of 33 targets, **27 have trivial geometric automorphism group**. The other six, all with trace -35, have a V4 group stable under Frobenius, cycling its three involutions. Their parameter codes are 13504, 28267, 60803, 74777, 149517 and 224622. This is the published degree-two elliptic-quotient route of Nart–Ritzenthaler, 2009, not a new construction. Quotient maps have not been evaluated here. All three trace +35 targets have trivial geometric automorphism group, hence admit no separable degree-two elliptic quotient. This does not exclude higher-degree maps or establish a new attack.

A derived sufficient CM test

The full written proof gives the following test. For a simple ordinary cubic restriction, a rational transverse isotropic ell-kernel, and geometric CM field $\mathbb{Q}(\sqrt{-\delta})$ with delta positive squarefree, decomposability forces positive sorted a,b,c of sum ell, whose pair products are degrees of inter-factor isogenies, and

$$H = 2\ell abc - (ab)^2 - (ac)^2 - (bc)^2 = \delta z^2, \quad z \in \mathbb{Z}.$$

For traces +/-35 the fundamental Frobenius discriminant is -823. The written ordinary CM/Tate-module argument forces inter-factor degrees to be represented by $8x^2+3xy+26y^2$. At ell=53 only (1,26,26), (8,8,37), (8,13,32) survive the degree test. The first has H<0; the other two have H=71,680 and 103,360, neither divisible by 823. Thus the written criterion gives geometric indecomposability for every such transverse 53-quotient. Arithmetic Torelli selects a genus-three Jacobian for one source twist. Its degree is 53^3, coprime to the source primes 137 and 239. The abstract torsion certificates check all 53^3 combinations per source trace, but contain no E[53] points.

This quotient has **not** been identified with any of the 33 explicit quartics. Neither the arithmetic twist nor a quotient equation/map is selected by this argument. The prime scan covers only ell=2 mod 3 up to 53; 53 is not claimed minimal. The sum-37 control (8,13,16) has H=823 times 64 and correctly passes this necessary arithmetic condition.

The new Lean project proves four named arithmetic statements, with a transitive audit of five theorems including a generated equation lemma. Two named statements use no axioms; the other two use only propext, Classical.choice and Quot.sound. The proof map explicitly leaves the CM degree form, unbounded witness bounds, polarized geometry, simplicity and Torelli outside the formal proof. No sorry, custom claim axiom or native-decision trust is used. The repository gate passes on 96 research sources, five freshly rebuilt and 91 archive-bound.

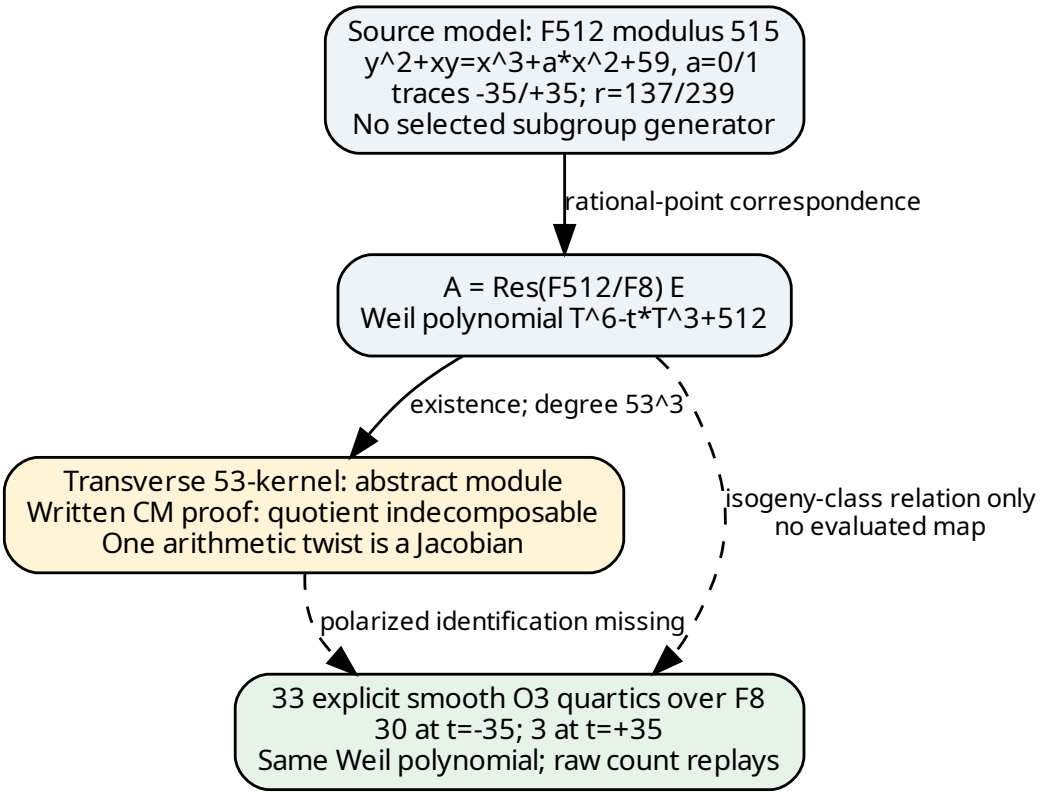


Figure 2: The computed equations and the abstract quotient are separate evidence. Dashed links denote missing identifications or evaluated maps.

Novelty, evidence and next step

Requested result	Current evidence
New recognition criterion with proof	Derived sufficient CM test; priority unresolved
New globally affected source family	Not established; only two new local trace classes
Explicit target equation	33 verified spectral targets in a published family
Explicit DLP transfer and recovered planted log	Missing
Lower genus or total attack cost than GHS/gGHS	Unmeasured; no claim

The full Tian 2023 journal priority comparison remains unfinished. Published Hess generalized descent already gives broad existence for these sources. The finite census is an observation, the CM implication is a written derivation, and the source-to-target map is an implementation gap. None is promoted to a novel vulnerability by absence from a local journal. The active problem specifies the next discriminating construction: identify a trace +35 target as the polarized quotient (or construct the actual target), evaluate the map on a declared r=239 subgroup, and independently recover a planted log.

Prospective contracts are QUARTIC_CUBIC_CHECKS.md, QUARTIC_AUTOMORPHISM_CHECKS.md, CUBIC_CM_CHECKS.md and CUBIC_CM_DISCRIMINANT_CHECKS.md. Code was frozen before each scientific run. Source commits, commands, raw outputs, hashes, failures and version transitions are in quartic-cubic-checks/artifact-manifest.json. All 48 Rust release tests and strict all-target Clippy pass. An earlier Clippy attempt failed on three style diagnostics; the corrected attempt is retained alongside it. The first corpus replay failed because the audit also found a generated equation theorem; the expectation was corrected to cover it, and the second replay passed. These are validation corrections, not changed scientific outcomes. Wall times are execution receipts only, without an isolated-host claim, cost calibration or rho comparison.

Canonical sources checked: ic-candidate-catalog/curves.yaml, ic-candidate-catalog/isogeny_routes.json, volcano-ic/figures/ and the existing GHS scope/kernel/cubic figures. No verified curve-graph edge or weak-curve catalogue entry is added: no map or subgroup transport exists. The earlier five-gluing overlap report remains valid for its separate 12-class scope. Frozen evidence is preserved; mutable global gates and the problem record receive explicit old/new hash transitions.